

Report

Sandanala Vinila

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Team Members:

M. Kiran Kumar(25VV1D8808)

S. Vinila(25VV1D8811)

Introduction:

The objective of this assignment is to build a Sequence-to-Sequence (Seq2Seq) model for transliteration using Recurrent Neural Networks. The model converts a Romanized Hindi word into its corresponding Hindi script representation. Hyperparameter tuning was performed using Weights & Biases Sweeps.

Dataset:

The Dakshina Hindi dataset was used for training and evaluation. The dataset contains pairs of Romanized Hindi words and their corresponding Hindi script words.

Example:

Romanized Input: amma

Hindi Output: अम्मा

Character-level tokenization was used to process the input and output sequences.

Model Architecture:

The model follows an Encoder-Decoder architecture consisting of:

1. Embedding Layer
2. Recurrent Layer (RNN/GRU/LSTM)
3. Decoder Network
4. Fully Connected Output Layer

The encoder processes the input sequence and the decoder generates the Telugu transliteration character by character.

Hyperparameter Sweep:

Weights & Biases Sweeps were used to explore different hyperparameter combinations.

The following parameters were varied:

1. Learning Rate
2. Hidden Size
3. Embedding Size
4. Number of Layers
5. Dropout
6. Cell Type (RNN, GRU, LSTM)

Multiple runs were executed to identify the best-performing configuration.

Results and Analysis:

The sweep results indicate that model performance improves with appropriate hidden dimensions and learning rates. LSTM and GRU models generally achieved better validation accuracy compared to simple RNNs.

The Validation Accuracy and Training Loss plots generated using W&B are shown below.

Section 1

Observations:

- 1.Validation accuracy increased steadily during training.
- 2.Training loss decreased across epochs, indicating successful learning.
- 3.LSTM-based models provided better transliteration performance.
- 4.Hyperparameter tuning significantly improved model accuracy.

Conclusion:

A Seq2Seq transliteration model was successfully implemented for Hindi transliteration. Weights & Biases Sweeps helped identify effective hyperparameter settings and improved overall model performance. The model was able to accurately convert Romanized Hindi words such as "amma" into their Hindi script equivalents.

<https://wandb.ai/sandalav-jawaharlal-nehru-technological-university/llm%20assignment/reports/Report--VmldzoxNzEwMzc4OA?accessToken=gl0oxt76a0ft6j6wwlt32d8zvtgoots3jnwpqu0pg8svcnt9bdxe2jbfg0x59mcw>